```python
from pyspark.sql.functions import col, sum

# Cast correct column
sales = sales.withColumn("total_amount", col("total_amount").cast("double"))
```

```python
from pyspark.sql.functions import desc

sales.groupBy("customer_id") \
     .agg(sum("total_amount").alias("total_amount")) \
     .orderBy(desc("total_amount")) \
     .limit(3) \
     .show()
```

Run Code

Follow for Spark tips & updates

TABLE | STDOUT | STDERR

```
+-----------+------------------+
|customer_id|      total_amount|
+-----------+------------------+
|          4|135.45999999999998|
|         22|            119.96|
|          2|            114.97|
+-----------+------------------+
```

Help Us Improve

```python
customers = spark.read.option("header", "true").csv("/samples/customers.csv"

customers.columns
sales = spark.read.option("header", "true").csv("/samples/sales.csv")

sales.columns
from pyspark.sql.functions import col, sum

# Cast correct column
sales = sales.withColumn("total_amount", col("total_amount").cast("double"))
customers.join(sales, "customer_id") \
        .groupBy("city") \
        .agg(sum("total_amount").alias("total_revenue")) \
        .show()
```

Run Code

Follow for Spark tips & updates

TABLE STDOUT STDERR

```
+-------------+-------------+
|         city|total_revenue|
+-------------+-------------+
|  Springfield|        69.97|
|      Oakland|         40.0|
|  San Antonio|        79.96|
| Philadelphia|         60.0|
| Indianapolis|        59.97|
```

Help Us Improv

```python
sales.columns
from pyspark.sql.functions import avg

sales.groupBy("customer_id") \
     .agg(avg("total_amount").alias("avg_amount")) \
     .show()
```

Run Code

Follow for Spark tips & updates

TABLE | STDOUT | STDERR

```
+-----------+----------+
|customer_id|avg_amount|
+-----------+----------+
|          7|    110.95|
|         15|      92.0|
|         11|      25.0|
|         29|     35.98|
|         42|      22.5|
|          3|     49.98|
|         30|     71.97|
|         34|      92.0|
|          8|      20.0|
|         22|    119.96|
|         28|      45.0|
|         16|      60.0|
|         35|      60.0|
|         43|      40.0|
```

Help Us Improve

```python
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("Test").getOrCreate()
customers = spark.read.option("header", "true").csv("/samples/customers.csv"

customers.columns
sales = spark.read.option("header", "true").csv("/samples/sales.csv")

sales.columns
#sort customers by total spend
from pyspark.sql.functions import sum, desc

sales.groupBy("customer_id") \
     .agg(sum("total_amount").alias("total_amount")) \
```

Run Code

Follow for Spark tips & updates

TABLE | STDOUT | STDERR

```
+-----------+------------------+
|customer_id|      total_amount|
+-----------+------------------+
|          4|135.45999999999998|
|         22|            119.96|
|          2|            114.97|
|          7|            110.95|
|         45|            109.96|
|         24|             99.95|
|         15|              92.0|
```

Help Us Improv

```python
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("Test").getOrCreate()
customers = spark.read.option("header", "true").csv("/samples/customers.csv"

customers.columns
sales = spark.read.option("header", "true").csv("/samples/sales.csv")

sales.columns
#avg order amount per customer
from pyspark.sql.functions import avg

sales.groupBy("customer_id") \
     .agg(avg("total_amount").alias("avg_amount")) \
```

Run Code

Follow for Spark tips & updates

TABLE STDOUT STDERR

```
+-----------+----------+
|customer_id|avg_amount|
+-----------+----------+
|          7|    110.95|
|         15|      92.0|
|         11|      25.0|
|         29|     35.98|
|         42|      22.5|
|          3|     49.98|
|         30|     71.97|
```

Help Us Improv

```python
from pyspark.sql import SparkSession

spark = SparkSession.builder.appName("Test").getOrCreate()
customers = spark.read.option("header", "true").csv("/samples/customers.csv"

customers.columns
sales = spark.read.option("header", "true").csv("/samples/sales.csv")
sales.columns
#city wise total revenue
from pyspark.sql.functions import sum
customers.join(sales, "customer_id") \
        .groupBy("city") \
        .agg(sum("total_amount").alias("total_revenue")) \
        .show()
```

Run Code

Follow for Spark tips & updates

TABLE | STDOUT | STDERR

```
+-------------+-------------+
|         city|total_revenue|
+-------------+-------------+
|  Springfield|        69.97|
|      Oakland|         40.0|
|  San Antonio|        79.96|
| Philadelphia|         60.0|
| Indianapolis|        59.97|
|    Nashville|        67.47|
|    San Diego|         95.5|
```

Help Us Improv